Here $(x_1, y_1) = (2, 4)$ and

$(x_2, y_2) = (3, -2)$

$\therefore (x-2)(x-3)+(y-4)(y+2) = 0$

$x^2+y^2-5x-2y-2 = 0$

3.4.3 General equation of a circle

The general equation of circle is $x^2+y^2+2gx+2fy+c = 0$ where g, f and c are constants.

i.e $x^2+y^2+2gx+2fy = -c$

$x^2+2gx+g^2-g^2+y^2+2fy+f^2-f^2 = -c$

$(x+g)^2-g^2+(y+f)^2-f^2 = -c$

$(x+g)^2+(y+f)^2 = g^2+f^2-c$

$[x-(-g)]^2+[y-(-f)]^2 = \left[\sqrt{g^2+f^2-c}\right]^2$

Comparing this with the circle $(x-h)^2+(y-k)^2 = r^2$

We get, centre is $(-g, -f)$ and radius is $\sqrt{g^2+f^2-c}$

NOTE

The general second degree equation $ax^2+by^2+2hxy+2gx+2fy+c = 0$ represents a circle if

(i) $a = b$ i.e., co-efficient of x^2 = co-efficient of y^2.

(ii) $h = 0$ i.e., no xy term.

Example 3.20

Find the centre and radius of the circle $x^2+y^2-8x+6y-24 = 0$

Solution

Equation of circle is

$x^2+y^2-8x+6y-24 = 0$

Here $g = -4$, $f = 3$ and $c = -24$

Centre $= C(-g, -f) = C(4, -3)$ and

Radius: $r = \sqrt{g^2+f^2-c}$

$= \sqrt{16+9+24} = 7$ unit.

Example 3.21

For what values of a and b does the equation

$(a-2)x^2+by^2+(b-2)xy+4x+4y-1 = 0$ represents a circle? Write down the resulting equation of the circle.

Solution

The given equation is

$(a-2)x^2+by^2+(b-2)xy+4x+4y-1 = 0$

As per conditions noted above,

(i) coefficient of $xy = 0 \Rightarrow b-2 = 0$

$\therefore b = 2$

(ii) coefficient of x^2 = Coefficient of y^2

$\Rightarrow a - 2 = b$

$a - 2 = 2 \Rightarrow a = 4$

$\therefore$ Resulting equation of circle is $2x^2+2y^2+4x+4y-1 = 0$

Example 3.22

If the equation of a circle $x^2+y^2+ax+by = 0$ passing through the points (1, 2) and (1, 1), find the values of a and b

Solution

The circle $x^2+y^2+ax+by = 0$ passing through (1, 2) and (1, 1)

$\therefore$ We have $1 + 4 + a + 2b = 0$ and $1 + 1 + a + b = 0$

$\Rightarrow a+2b = -5$ (1)

and $a+b = -2$ (2)

Solving (1) and (2), we get $a = 1$, $b, = -3$.

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